CO2 - AT3

Class Test

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Course code : ITA 0502

Course Name : computer Vision

1) Suitable technique :- log transformation

Processing steps :-

1. Read the dark image & convert into grayscale
2. Apply the log transformation using $s = (\log(1+r))$ where r is the input intensity & c constant
3. Using $s = cr^{\gamma}$, where $\gamma < 1$ brightness
4. Output intensity values range 0-255
5. Display the enhanced image and compare it with the original image.

2) Histogram Equalization for fingerprint Enhancement technique :- Histogram Equalization

* convert the fingerprint image into grayscale
* Calculate the histogram to determine the frequency of each intensity level
* Map each original intensity using CDF to redistribute pixel values

Result :- Contrast improves, Making fingerprint

3) Salt and pepper Noise Removal

Technique :- Median filtering

* Read the noisy image
* Select a small window, such as 3x3
* Sort the pixel with the Median value.

Justification :-

unlike averaging filters the median filter removes Extreme black and white noise values while preserving edges.

4) Image sharpening for PCB Trace Inspection

Technique :- laplacian sharpening

Read the blurred PCB image and convert it to grayscale

Apply laplacian operator to delete rapid intensity changes & edges

Image using $g = f - \nabla^2 f$, with the sign depending on the kernel convention.

Domain low pass filtering

Technique :- Gaussian low-pass filter

* Convert the drone image into grayscale
* Apply the fourier transform
* Multiply the frequency spectrum by Gaussian low pass filter to suppress

* Cutoff frequency :-

A lower cutoff removes more noise but also blurs details.

6) High-pass filtering for retinal Blood vessels

Technique :- High-Boost filtering

* Read the retinal fundus image & convert it to grayscale.
* Apply a low pass filter
* Add a scaled version of these details to original image using $g = f + k(f - f_{blur})$
* where $k > 0$

Result :- Tini Blood vessels and fine edges.

7) point and line Detection for Bottle inspection

Technique :- point detection and directional line marks

* Convert the bottle image to grayscale
* Apply a point detection mask

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

to identify isolated intensity.

8) Thresholding for Handwritten Answer sheets

Technique :- Adaptive thresholding

* Read the Answer-sheet image
* Divide the image into small local neighbours
* Calculate a local pixel threshold values

Justification :-

Global thresholding users one threshold for the entire image.

9) Region Growing for late Extraction

Technique :- Region Growing

Read the satelite image and select a seed pixel. inside the take

Examine neighbouring pixels around the Add neighbouring pixels whose they intensity or colour is sufficiently similar to the region.

10) Edge linking & Boundary Detection for mechanical Gears

Technique :- Edge linking using thaugh transform

* Convert the gear image to grayscale and reduce noise using suitable filter
* Trace the connected edges to dotain the complete gear outline

Result :- Broken edge segments are linked, Producing a more continuous geag